Lab Exercise 16- Creating a ConfigMap from a Folder

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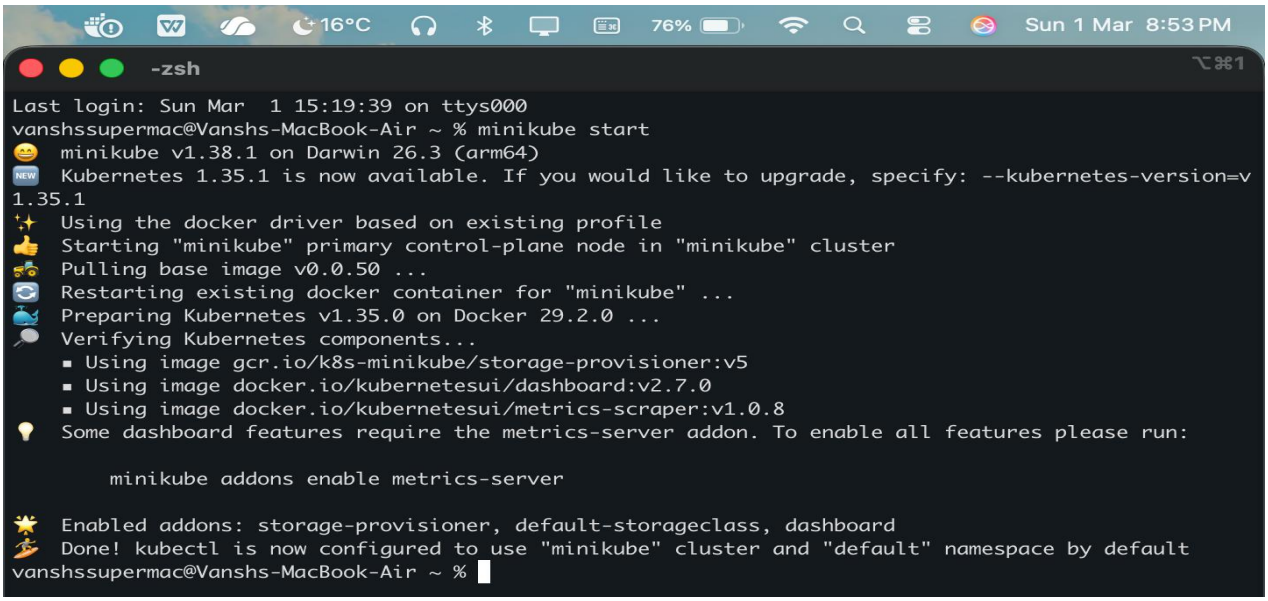
In this lab exercise, you will learn how to create a ConfigMap in Kubernetes from a folder containing multiple files. This allows you to manage multiple configuration files as a single ConfigMap.

Step 1: Set Up Kubernetes Cluster

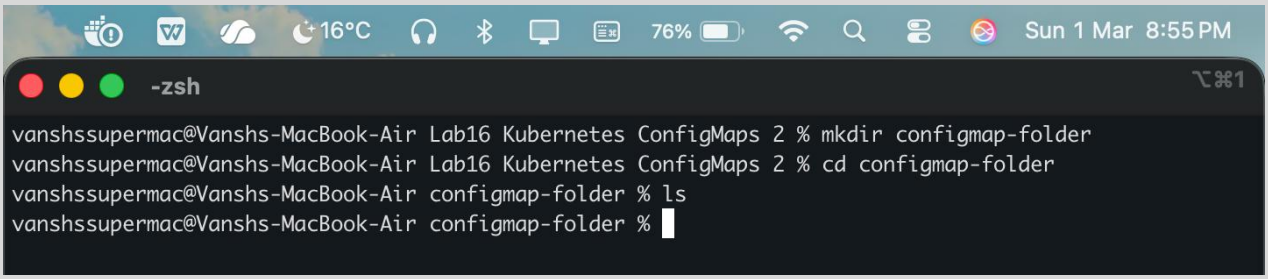
Ensure you have access to a Kubernetes cluster. You can use a local setup with Minikube, kind, or use a cloud-based Kubernetes service.



Step 2: Prepare the Folder with Multiple Files

Create a directory named configmap-folder:

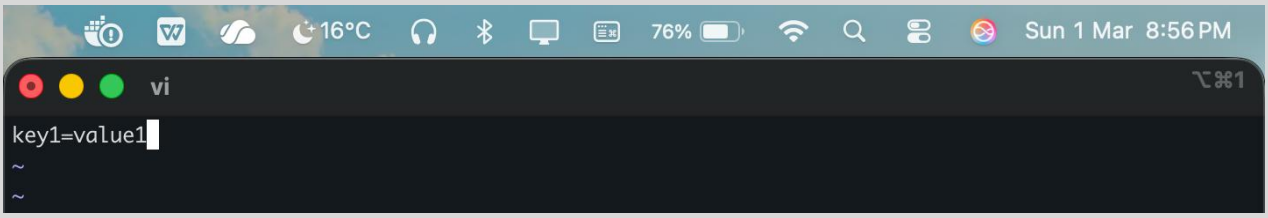
```
mkdir configmap-folder
cd configmap-folder/
```



Create multiple configuration files inside the directory:

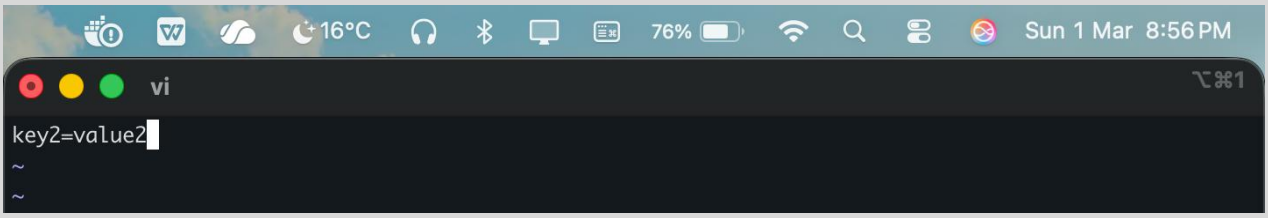
config1.txt with content:

```
key1=value1
```



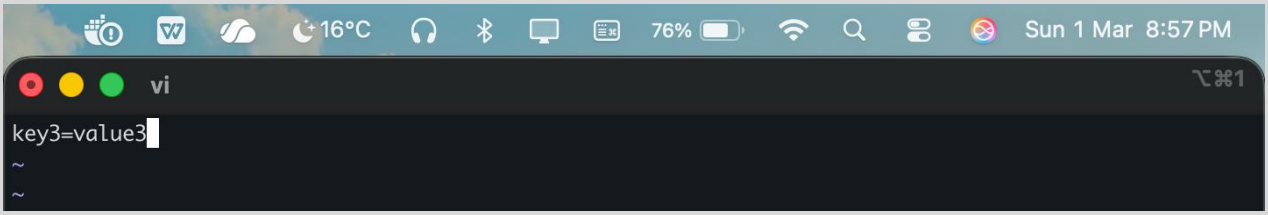
config2.txt with content:

```
key2=value2
```



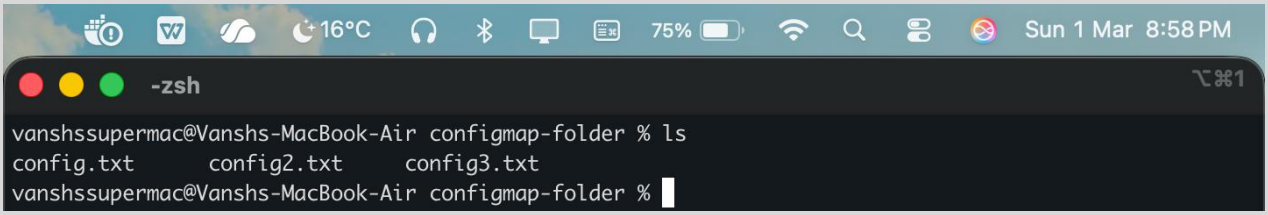
config3.txt with content:

```
key3=value3
```



Verify the directory structure:

```
ls configmap-folder/
cd ..
```

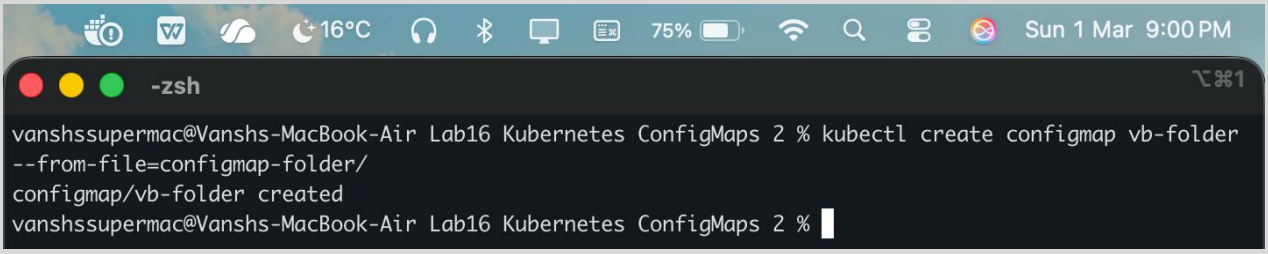


You should see the files config1.txt, config2.txt, and config3.txt.

Step 3: Create a ConfigMap from the Folder

Create a ConfigMap named folder-configmap from the folder:

```
kubectl create configmap folder-configmap --from-file=configmap-folder/
```



Verify the ConfigMap:

kubectl get configmap folder-configmap -o yaml

```
vanshssupermac@Vanshs-MacBook-Air Lab16 Kubernetes ConfigMaps 2 % kubectl get configmap vb-folder -o
yaml
apiVersion: v1
data:
  config.txt: |
    key1=value1
  config2.txt: |
    key2=value2
  config3.txt: |
    key3=value3
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-01T15:30:52Z"
  name: vb-folder
  namespace: default
  resourceVersion: "10990"
  uid: 0bba203c-9b98-4ba2-b61d-5540f7b29f40
vanshssupermac@Vanshs-MacBook-Air Lab16 Kubernetes ConfigMaps 2 %
```

You should see the folder-configmap with the contents of all the files in the configmap-folder.

Step 4: Use the ConfigMap in a Pod

Create a file named **pod-folder-configmap.yaml** with the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers:
  - name: nginx
    image: nginx
    envFrom:
    - configMapRef:
```

```
    name: folder-configmap
```

```
apiVersion: v1
kind: Pod
metadata:
  name: configmap-pod
spec:
  containers
  - name: nginx
   image: nginx
   envFrom:
   - configMapRef:
     name: folder-configmap
~
~
```

In this manifest:

- The Pod mounts the folder-configmap at /etc/config.
- The busybox container concatenates and prints the contents of the files.

Apply the manifest to create the Pod:

```
kubectl apply -f pod-folder-configmap.yaml
```

```
vanshssupermac@Vanshs-MacBook-Air Lab16 Kubernetes ConfigMaps 2 % kubectl apply -f vb-pfc.yaml
pod/configmap-pod created
vanshssupermac@Vanshs-MacBook-Air Lab16 Kubernetes ConfigMaps 2 %
```

Check the status of the Pod:

```
kubectl get pods
```

```
vanshssupermac@Vanshs-MacBook-Air Lab16 Kubernetes ConfigMaps 2 % kubectl get pods
NAME            READY   STATUS                       RESTARTS   AGE
configmap-pod   0/1     CreateContainerConfigError   0          28s
vanshssupermac@Vanshs-MacBook-Air Lab16 Kubernetes ConfigMaps 2 %
```

View the Pod logs to see the content of the mounted files:

```
kubectl exec -it configmap-pod -- printenv
```

You should see the contents of config1.txt, config2.txt, and config3.txt printed.

Step 5: Clean Up

After completing the exercise, clean up the resources created:

```
kubectl delete pod folder-configmap-pod
kubectl delete configmap folder-configmap
rm -r configmap-folder
```

```
vanshssupermac@Vanshs-MacBook-Air Lab16 Kubernetes ConfigMaps 2 % rm -r configmap-folder
vanshssupermac@Vanshs-MacBook-Air Lab16 Kubernetes ConfigMaps 2 %
```

Thank You